

lab_07

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```
library(tidyr)
library(ggplot2)
url <- "https://tinyurl.com/UK-foods"
food_df <- read.csv(url, row.names=1)
dim(food_df)
```

```
[1] 17  4
```

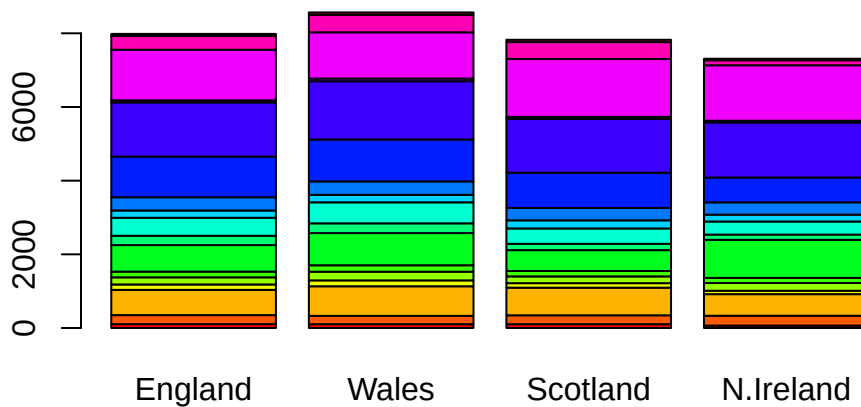
Q1. How many rows and columns are in your new data frame named x? What R functions could you use to answer this questions?

There are 17 rows and 4 columns

Q2. Which approach to solving the 'row-names problem' mentioned above do you prefer and why? Is one approach more robust than another under certain circumstances?

I'd rather use row.names=1 since it avoids extra manipulation of the dataset

```
barplot(as.matrix(food_df), beside=F, col=rainbow(nrow(food_df)))
```



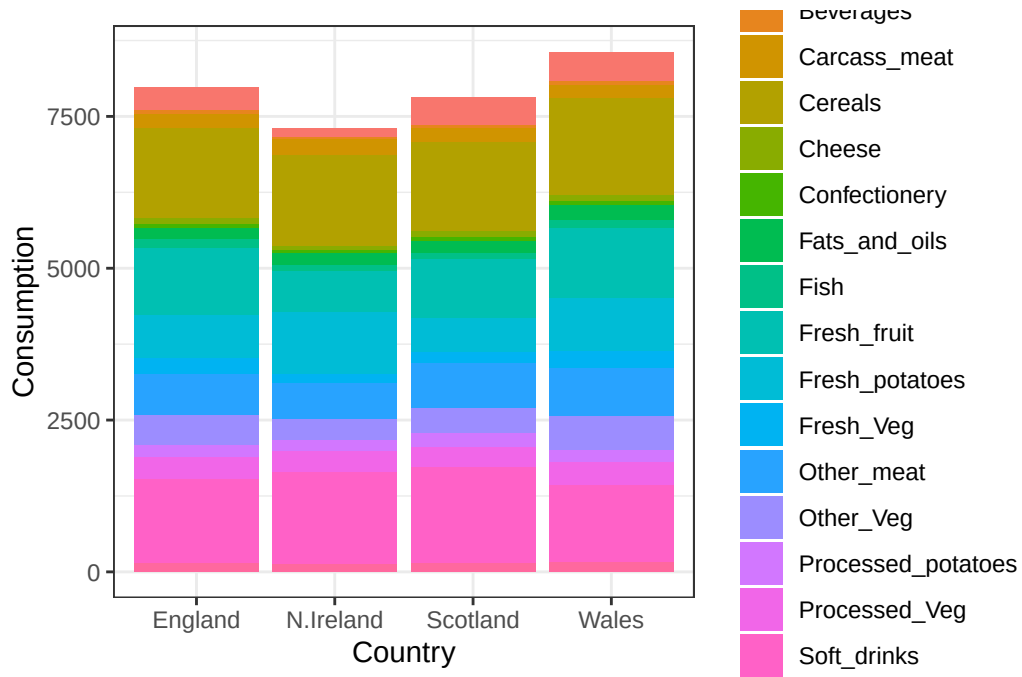
Q3. Changing what optional argument in the above `barplot()` function results in the following plot?

`beside=T`

```
x_long <- food_df |>
  tibble::rownames_to_column("Food") |>
  pivot_longer(cols = -Food,
               names_to = "Country",
               values_to = "Consumption")
head(x_long)
```

```
# A tibble: 6 x 3
  Food          Country Consumption
<chr>         <chr>         <int>
1 "Cheese"      England          105
2 "Cheese"      Wales            103
3 "Cheese"      Scotland         103
4 "Cheese"      N.Ireland         66
5 "Carcass_meat " England          245
6 "Carcass_meat " Wales            227
```

```
ggplot(x_long) +
  aes(x = Country, y = Consumption, fill = Food) +
  geom_col(position = "stack") +
  theme_bw()
```



Q4. Changing what optional argument in the above `ggplot()` code results in a stacked barplot figure?

`geom_col(position = "stack")`

Q7. Complete the code below to generate a plot of PC1 vs PC2. The second line adds text labels over the data points.

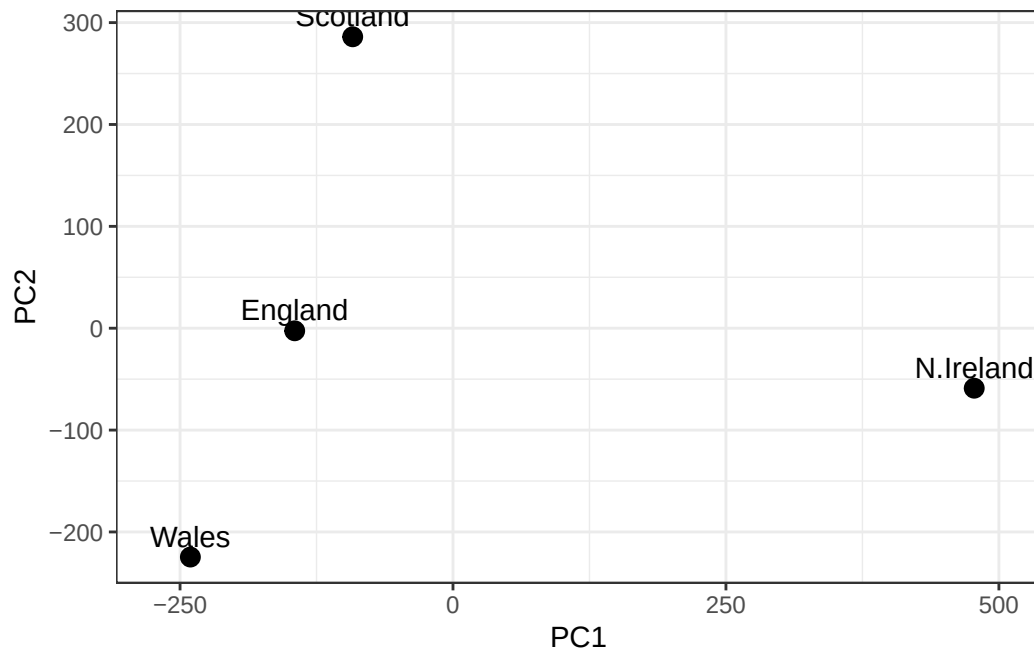
```
pca <- prcomp(t(food_df))
# Create a data frame for plotting
df <- as.data.frame(pca$x)
df$Country <- rownames(df)

ggplot(pca$x) +
  aes(x = PC1, y = PC2, label = rownames(pca$x)) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5) +
  xlim(-270, 500) +
```

```

xlab("PC1") +
ylab("PC2") +
theme_bw()

```

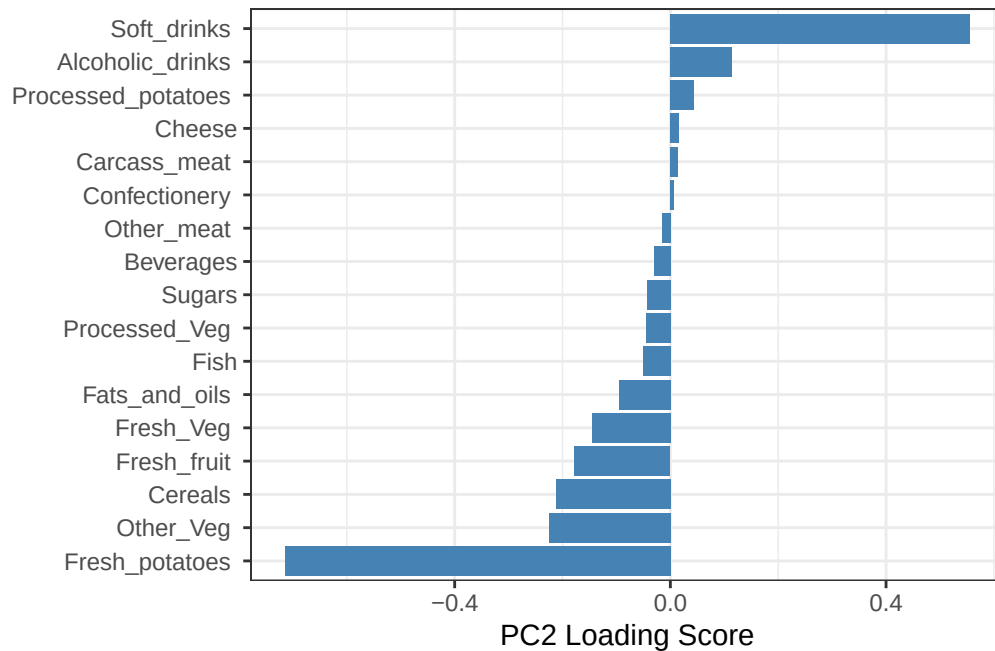


Q9: Generate a similar 'loadings plot' for PC2. What two food groups feature prominently and what does PC2 mainly tell us about?

```

ggplot(pca$rotation) +
  aes(x = PC2,
       y = reorder(rownames(pca$rotation), PC2)) +
  geom_col(fill = "steelblue") +
  xlab("PC2 Loading Score") +
  ylab("") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 9))

```



```
url2 <- "https://tinyurl.com/expression-CSV"
rna.data <- read.csv(url2, row.names=1)
dim(rna.data)
```

```
[1] 100 10
```

Q10: How many genes and samples are in this data set? How many PCs do you think it will take to have a useful overview of this data set (see below)?

There are 100 genes in the dataset